

Sara Mascia

PhD, Curriculum Vitae

Research Interests

- **General:** Astronomy outreach
- **Astronomical instrumentation, methods and techniques:** Telescopes (ultraviolet, optical, infrared), photometry, spectroscopy
- **Galaxies:** active, distances and redshifts, evolution, high-redshift
- **Cosmology:** observations, early Universe

Education

- 2024 – Present **ISTA Prize Fellow**, affiliated with the Matthee group, Institute of Science and Technology Austria (ISTA)
- 2021 – 2024 **PhD in Astronomy, Astrophysics and Space Science**, University of Rome "Tor Vergata" with the collaboration of the University of Rome "Sapienza" and the National Institute of Astrophysics (INAF) + **Amaldi Research Center (ARC) fellow**
- **PhD Thesis:** "Tracing the properties of Lyman continuum emitting galaxies over cosmic epochs" under supervision of **Prof. Laura Pentericci** and **Dr. Marco Castellano**
- 2019 – 2021 **M.Sc. cum Laude in Astrophysics and Space Physics**, LM-58, University of Milano Bicocca, Italy
- **M.Sc. Thesis:** "Star formation and quenching of low mass galaxies at $z \sim 1 - 2$ in the MUSE Ultra Deep Field (MUDF) survey", under supervision of **Prof. Michele Fumagalli** and **Dr. Matteo Fossati**
- 2016–2019 **B.Sc. cum Laude in Physics**, L-30, Università degli Studi di Cagliari, Italy
- **B.Sc. Thesis:** "Analisi temporale del pulsatore X al millisecondo XTE J0929-314", under supervision of **Prof. Alessandro Riggio**

International schools attended

- Oct. 2023 **4th Italian Astrostatistics School**, INAF - Osservatorio Astronomico di Brera, Milano
- Sept. 2023 **18th Heidelberg Summer School on Unraveling Galaxy Evolution with JWST**, Max Planck Research School for Astronomy and Cosmic Physics at the University of Heidelberg (IMPRS-HD)
- May 2022 **IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution**, Instituto de Astrofísica de Andalucía headquarters (IAA-CSIC), Granada
- Feb. 2022 **DAWN Winter School in Astrophysics**, University of Copenhagen, Faculty of Science

International collaborations

- 2021 – Present **VANDELS**, (PID 194.A-2003, PI L. Pentericci and R. McLure)
- 2022 – Present **GLASS-JWST-ERS**, (JWST-ERS-1324: PI T. Treu)
- 2022 – Present **FLAGS**, (First Light And Galaxy aSsembly, PI: L. Pentericci and R. Schneider)

- 2023 – Present **CEERS**, (*Cosmic Evolution Early Release Science survey, JWST-ERS-1345, PI: S. Finkelstein*)
- 2023 – Present **NGDEEP**, (*The Next Generation Deep Extragalactic Exploratory Public, JWST-GO-2079, PI: S. Finkelstein*)
- 2024 – Present **LyC22**, (*Deep spectroscopic insights on star-forming galaxies 2.2 Gyr after the Big Bang, JWST-GO-1869, PI: D. Schaerer*)

Observational programs

- Co-I of **To catch an AGN: Ionizing Emission and AGN Activity of COLA1 in the Epoch of Reionization**, GO 7722, PIs A. Torralba, J. Matthee (JWST)
- Co-I of **Searching for Population III stars around massive interacting galaxies in the reionization epoch**, GO 7014, PI A. Calabro' (JWST)
- Co-I of **A deep look into the physics of the interstellar medium 360 Myr after the Big Bang**, GO 7201, PI M. Castellano (JWST)
- Co-I of **The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS)**, PI M. Dickinson (JWST)
- Co-I of **On the nature of UV emission line galaxies: a study of CIV emitters at the Cosmic Noon**, PI A. Saldana Lopez (XSHOOTER)
- Co-I of **Connecting Lyman Continuum emission to UV spectral properties in VANDELS galaxies at $z \sim 3.5$** , PI L. Pentericci (MODS-LBT)
- Co-I of **Spectroscopic confirmation of an unexpected population of bright galaxies at cosmic dawn**, GO 3073, PI M. Castellano (JWST)

Miscellaneous

- 2025 **A&A Award for Outstanding PhD Article**
- March 2024 **Visitor at Observatoire de Genève, Versoix, Switzerland**, Reduction and analysis of JWST/NIRSpec data. Collaboration with prof D. Schaerer.
- 2023 – Present **LBT Proposal Science Reviewer**
- Nov. – Dec. 2022 **Visitor at UCLA, Los Angeles, CA, USA**, Acquirement of skills in reduction and analysis of JWST/NIRSpec data. Collaboration with prof T. Treu and PhD. G. Roberts Borsani.

Conferences

- **Invited talk:** *A (not so) Complete Unknown: multiple predictors of Lyman Continuum escape in the early Universe*, Conference: Escape of Lyman radiation from galactic labyrinths, Kolymbari, Crete, Greece, 8-11 April 2025
- **Contributed talk:** *Understanding Lyman Continuum escape in high-redshift galaxies: insights from JWST observations*, Conference: Understanding the Epoch of Reionization II, Sesto, Val Pusteria, Italy, 13-17 January 2025
- **Contributed talk:** *The impact of mergers on the Lyman Continuum photons escape in Galaxies during the epoch of reionization*, Conference: Cosmic Dawn at High Latitudes Conference, Stockholm, Sweden, 24-28 June 2024
- **Invited talk:** *Unveiling the Cosmic Dawn: Exploring the High-Redshift Universe with JWST*, Physics Colloquium, Università di Cagliari, INFN, Sezione di Cagliari, Italy, 27 March 2024
- **Contributed talk:** *Escaping the labyrinth of reionization with JWST observations*, Conference: The growth of galaxies in the Early Universe - XIV, Sesto, Val Pusteria, Italy, 22-26 January 2024

- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Conference: 12th Young Researcher Meeting, Monte Porzio Catone, Italy, 2-4 October 2023
- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Conference: The James Webb Space Telescope turns one: the birth and growth of galaxies, Sesto, Val Pusteria, Italy, 10-14 July 2023
- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Conference: Shedding new light on the first billion years of the Universe, Marseille, France, 3-7 July 2023
- **Contributed talk:** *The indirect indicators' thread: escaping the labyrinth of reionization*, Students' Day @ OAR, Roma, Italy, 9 May 2023
- **Contributed talk:** *Reionization's labyrinth: how to escape following the indirect indicators' thread*, Conference: Escape of Lyman radiation from galactic labyrinths, Kolymbari, Crete, Greece, 18-21 April 2023
- **Contributed talk:** *Closing in on the sources of cosmic reionization: first results from the GLASS-JWST program*, Conference: The growth of galaxies in the Early Universe - VIII, Sesto, Val Pusteria, Italy, 13-17 March 2023
- **Contributed talk:** *Insight into the reionization epoch from cosmic high-noon studies of UV emission galaxies*, European Astronomical Society Annual Meeting, Valencia, Spain, 27 June – 1 July 2022
- **Contributed talk:** *The VANDELS Survey: Properties of the CIV $\lambda 1550$ emitters at $z \sim 2.5 - 5$* , VANDELS final meeting, Bologna, Italy, 12-13 May 2022
- **Contributed talk:** *The VANDELS Survey: Properties of the CIV $\lambda 1550$ emitters at $z \sim 2.5 - 5$* , Students' Day @ OAR, Roma, Italy, 10 May 2022

Outreach

- 01/12/2023 **Author of the article in the INAF magazine UNIVERSI, entitled "Ecco le galassie che hanno reionizzato l'universo" (Italian)**
- 29-30/09/2023 **Partecipation at the "Notte dei Ricercatori 2023", Città dell'altra economia, Testaccio, Roma**
- 18/03/2023 **Cosmo 2050, radio interview, Giornale Radio**
- 13/03/2023 **Ecco le galassie che hanno reionizzato l'universo, press release, Media INAF (Italian), + 7 other articles on different italian newspapers based on it**

Teaching and tutoring

- 2023-2024 **Representative of PhD Candidates, University of Rome "Tor Vergata" with the collaboration of the University of Rome "Sapienza" and the National Institute of Astrophysics (INAF)**
- 2023 **PCTO INAF: external tutor, Liceo Scientifico Statale Vito Volterra, Ciampino (RM)**

Skills

- **Scientific Software:** MATLAB, LaTeX, DS9, QFitsView, Overleaf, PyMultinest, Pandora, Sippi
- **Operating Systems:** Linux, Mac OS X, Windows
- **Programming Languages:** Python/Python3, C, C++
- **Languages:** Italian (native), English (fluent)